MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
Discipline Specific Elective (DSE) –MG5DSECHE300- QUANTUM MECHANICS, SPECTROSCOPY & GROUP THEORY
MODEL QUESTION PAPER
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions
Answer any 10 questions
Each question carries 2 marks

1. What would be the wavelength of a moving particle if its velocity is doubled? (CO-1,A)
2. Which of the following is an eigen function of d/dx a) x^2 b) e^{ax} . Why? (CO-1,An)
3. Give time independent Schrodinger equation for a particle in one-dimensional box of infinite potential wall. (CO-1, U)
4. Explain different regions of electromagnetic radiation used in spectroscopy (CO-2,U)
5. What is meant by a rigid rotor ? (CO-2,K)
6. Define zero-point energy in vibrational spectroscopy. (CO-2,K)
7. Show how many spin states are possible for Carbon-13? (CO-3, U)
8. Define chemical shift. (CO-3, K)
9. State the Laporte selection rule for electronic transitions. (CO-3, K)
10. Distinguish between C_{nv} and C_{nh} point groups. (CO-4, U)
11. Explain the term classes in group theory with an example. (CO-4, U)
12. Give an example of a molecule belonging to D_3h point group and identify its symmetry Elements. (CO-4, U)

[10 × 2 = 20]

Part B

Short Answer Questions
Answer any 6 questions
Each question carries 5 marks

13. Discuss the limitations of classical mechanics with reference to any two of the following phenomena a) Blackbody radiation b) Photoelectric effect c) Compton effect. (CO-1, U)

14. What is meant by separation of variables in solving Hydrogen atom wave equation?

(CO-1, A)

15. Distinguish between natural line broadening and Doppler broadening. (CO-2, U)

16. Compare an overtone band with a fundamental vibrational band. (CO-2, U)
17. A molecule contains a carbonyl group (C=O). Identify the molecular orbitals and molecular transitions involved. (CO-3, A)
18. Write a short note on coupling constant. (CO-3, U)
19. Write a note on point groups to which molecules of high symmetry (MHS) belong. (CO-4, U)
20. Write down the matrix representation for centre of symmetry(i)and identity (E). (CO-4, U)

[6 × 5 = 30]

Part C

Short Essay Questions
Answer any 2 questions
Each question carries 10 marks

21. Discuss the radial probability distribution curves of electrons in hydrogen atom. (CO-1, U)
22. Explain the concept of anharmonicity in a diatomic molecule and explain the morse potential represents its anharmonic vibrational behaviour. (CO-2,U)
23. Outline briefly the factors affecting chemical shift. (CO-3, U)
24. S_n operation generates n distinct operations when n is even and 2n operations when n is odd. Justify the statement with respect to S_3 and S_6 operations. (CO-4, A)

[2 × 10 = 20]
